NSC2282: Biodiversity and Nature Conservation

Academic Essay

Portfolio note

This essay grew out of a reflective assignment for Biodiversity and Nature Conservation, but I have rebuilt it as an argument-driven academic essay rather than a personal reflection. It draws on reporting I did for the course — an interview with senior climate journalist Mostofa Yusuf of Prothom Alo — alongside peer-reviewed and documentary sources. Every figure has been checked against a named source and attributed; where estimates differ across sources, the range is reported rather than a single convenient number.

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1. The accounting problem

A mangrove forest does work that never appears on a balance sheet. It buffers storm surges, stores carbon, filters pollutants, holds soil against erosion, and shelters the species that keep a coastal ecology functioning. None of this is invoiced. When a government weighs a forest against an industrial zone, the zone arrives with a projected return measured in dollars and the forest arrives with a value of, formally, nothing. The asymmetry is not an oversight. It is built into how development decisions are costed, and it is the reason forests tend to lose.

This essay argues that Bangladesh's pattern of development-driven deforestation is best understood not as a series of unfortunate local decisions but as the predictable output of an accounting system that cannot see ecological value. The argument runs in three steps. First, the concept of ecosystem services explains why the value of nature is real but invisible to conventional cost-benefit analysis. Second, the clearance of the Mirsarai mangroves and the slow disappearance of Dhaka's urban trees show that same invisibility producing decisions at two very different scales. Third, the recent attempt to put a monetary figure on Dhaka's trees — and the courtroom challenge to the Mirsarai project — suggest both the promise and the limits of trying to make the uncounted count.

2. Why nature is valuable and invisible at once

The framework of ecosystem services, formalised by the Millennium Ecosystem Assessment (2005), classifies the benefits people derive from ecosystems into provisioning services such as food and timber, regulating services such as climate stabilisation and flood control, supporting services such as nutrient cycling, and cultural services such as recreation and meaning. The conceptual move that matters here is the recognition that most of these services are non-market goods: they are consumed by everyone, owned by no one, and priced nowhere. Costanza and colleagues (1997) made the point unforgettable by attempting to put a global figure on them, estimating the annual value of the world's ecosystem services at a sum on the order of the entire global economy — not because anyone would or could pay it, but to dramatise how large a value sits entirely outside the market.

The problem this creates for decision-making is structural. Standard cost-benefit analysis compares priced quantities. A development project generates revenues, employment, and tax that can be counted in the year they arrive. The forest it replaces generated services that were never counted, so their loss registers as zero rather than as a cost. The result is a systematic bias: projects that convert non-market natural capital into market output look more profitable than they are, because the analysis subtracts nothing for what is destroyed. This is what economists call an externality, and biodiversity loss is one of the largest uncosted externalities in the development process (TEEB, 2010). The senior journalist I interviewed for this essay put the same idea in plainer language: an economic zone shows you a measurable return in one year, while the clean air a forest provides is something “we don’t even count until we are lying in a hospital bed” (M. Yusuf, personal communication, August 2025).

The augmentation of this critique by Bangladeshi reality is severe, because the country starts from a position of acute scarcity. Per-capita green coverage in Bangladesh sits far below international benchmarks; the United Nations Environment Programme suggests roughly a quarter of a city’s area should be green, while Dhaka’s tree cover is under eleven percent (BFD & USFS, 2024). When the baseline stock of natural capital is already this thin, the marginal value of each remaining hectare is correspondingly higher — which is precisely the value the accounting cannot see.

3. The macro case: clearing the Mirsarai mangroves

The clearest illustration of the accounting failure at scale is the Mirsarai Economic Zone in Chattogram, since renamed the National Special Economic Zone. Developed from 2015 on a stretch of reserved forest, the project cleared a vast area of coastal mangrove. The exact toll is reported differently across sources, and the divergence is itself informative: The Daily Star reported the felling of 5.2 million trees across roughly 30,000 acres (The Daily Star, 2023), while investigative reporting in The Business Standard and subsequent coverage put the figure far higher, at more than 55 million trees cleared since 2015 across 22,335 acres of reserved forest, including hundreds of acres of mangrove (The Business Standard, 2025; The Climate Watch, 2025). What no source disputes is that an ecosystem that had taken decades to mature — home, by one account, to thousands of spotted deer and more than a hundred bird species — was converted to industrial land.

The journalist I interviewed had covered these projects directly, and his assessment was blunt: the last fifteen years had been “marked by the fall of trees,” with mountains, hills, and forests cleared in the name of development, and trees “always sacrificed first, usually for economic gain” (M. Yusuf, personal communication, August 2025). His frustration was not with development as such but with the refusal to pursue it without destruction — the routine choice of the shortcut where vacant private land or an accommodated plan could have spared the forest. A Forest Department officer made the same point on the record, noting that the zone could have been built on available private land, as other zones had been (The Business Standard, 2025).

The mitigation story confirms the diagnosis. On paper, such projects pledge compensatory afforestation; BEZA spoke of planting two million saplings to offset the Mirsarai clearance (The Daily Star, 2019). But replanting saplings does not reconstitute a mature mangrove ecology, and the journalist’s judgment — that you cannot replace a decades-old forest sustaining a balanced fauna by planting a few hundred or even a few thousand trees — is consistent with the ecological literature on the difficulty of restoring complex ecosystems. The lost forest also represented forgone income that the accounting never registered: based on European carbon-market rates, one estimate held that the Mirsarai mangroves alone could have generated at least 16 million dollars a year in carbon-credit revenue (The Business Standard, 2025). Even on the market’s own terms, in other words, the destruction was not obviously profitable; it only looked profitable because the carbon value was uncounted.

4. The micro case: the disappearing trees of Dhaka

The same logic operates inside the city, more quietly. Dhaka’s first comprehensive tree census, the Urban Tree Inventory conducted by the Bangladesh

Forest Department with the United States Forest Service and now published in the peer-reviewed literature, found roughly 1.3 million trees across the city — about one tree for every seven residents — with tree cover below eleven percent, far under what a tropical city of Dhaka's latitude could sustain (Khan et al., 2026; BFD & USFS, 2024). The inventory recorded 110 species across 33 families, of which more than sixty percent are exotic rather than native, with mango, mahogany, and coconut the most common.

What makes the inventory analytically important is that it did precisely what the ecosystem-services framework prescribes: it used the i-Tree Eco modelling tool to attach a monetary figure to non-market services. The trees were estimated to remove on the order of 538 tons of air pollutants annually and to provide ecosystem benefits valued at roughly USD 6.7 million per year, while storing carbon worth a further USD 3.2 million — figures reported locally in the equivalent Bangladeshi-taka terms (Khan et al., 2026). For the first time, the value that conventional accounting treated as zero had a number attached to it. The pollutant-removal service alone, the inventory noted, would cost a substantial sum to replicate by artificial means.

Dhaka urban forest (first inventory)	Finding
Total trees	~1.3 million (about 1 per 7 residents)
Tree cover	Below 11% (vs UNEP guidance of ~25%)
Species / families	110 species across 33 families; 60%+ exotic
Most common species	Mango, mahogany, coconut
Annual pollutant removal	~538 tons/year
Estimated ecosystem benefit	~USD 6.7 million/year, plus ~USD 3.2 million stored carbon

Source: Khan et al. (2026), Scientific Reports; BFD & USFS (2024), Urban Tree Inventory of Dhaka City.

The longer trend the inventory now anchors is steep. Earlier remote-sensing work has reported Dhaka's green cover collapsing across recent decades, and planners point to designated green zones being absorbed by infrastructure backed by both public and private interests (Khan, as cited in Prothom Alo, 2025). The consequence is not only aesthetic. The loss of the city's lower vegetation layers — grass and shrubs, not only trees — removes the surfaces that moderate ground-level heat, and Dhaka now runs hotter than its surrounding districts as built mass replaces green mass (environmental scientists cited in Prothom Alo, 2025). The journalist I interviewed extended the point to public health, observing that there is a real and uncounted mortality and morbidity cost to heat exposure in a treeless city — heatstroke, ultraviolet exposure, and the chronic stress of an overheated environment — that policy debate barely acknowledges (M. Yusuf, personal communication, August 2025). These, too, are ecosystem services rendered visible only by their absence.

5. Making the uncounted count — promise and limit

If the diagnosis is an accounting failure, the obvious remedy is better accounting, and two recent developments gesture toward it. The Urban Tree Inventory's monetisation of Dhaka's tree services is one: once a forest's regulating services carry a defensible price, they can in principle enter the cost-benefit calculation that previously ignored them. The legal challenge to the Mirsarai zone is the other. Following an investigative report, the High Court issued a Rule Nisi questioning the project's legality under the Forest Act of 1927 and ordered a status quo on construction, finding a prima facie case that the development had caused serious damage to ecology and biodiversity (The Business Standard, 2025). Here the value of the forest was asserted not through a price but through a legal right — a different way of forcing the uncounted into the decision.

Both remedies have limits worth stating plainly. Monetary valuation can backfire: once a forest is priced, it becomes tradeable, and a sufficiently large development return can still "buy" the right to destroy it, which is part of why the carbon-offset and compensatory-afforestation mechanisms have so often produced saplings where mangroves stood. Valuation makes nature legible to the market, but legibility is not the same as protection. The legal route avoids that trap by treating forest as a protected category rather than a priced commodity, but it is slow, reactive, and dependent on the contingencies of litigation and political will — in the Mirsarai case, an intervention came only after most of the clearing was already done. Neither remedy resolves the deeper tension the course kept returning to: development and conservation both make legitimate claims, and the question is not whether to choose one but how to design decisions that do not treat the destruction of natural capital as free.

6. Conclusion

Bangladesh's deforestation is often narrated as a story of greed or weak governance, and those factors are real. But the more durable explanation is structural: a decision system that counts the returns of development and ignores the services of nature will clear forests not occasionally but reliably, because on its own measure the forest is worth nothing. The Mirsarai mangroves and Dhaka's vanishing trees are the same failure at two scales — the macro and the micro of an economy that cannot price what keeps it alive. The recent efforts to attach a number to Dhaka's trees and a legal shield to Mirsarai's forest are early attempts to repair the accounting, and they matter. But the lesson of the ecosystem-services literature is that the repair has to be structural rather than cosmetic. Until the loss of a forest registers as a cost in the calculation that authorises its clearance, the shortcut the journalist described — cut the forest, level the land, build the project — will keep being the rational choice on paper, and the wrong one in every other respect.

References

Bangladesh Forest Department & United States Forest Service. (2024). Urban tree inventory of Dhaka City. Bangladesh Forest Department; USFS Compass Program (USAID).

Costanza, R., d'Arge, R., de Groot, R., Farber, S., Grasso, M., Hannon, B., Limburg, K., Naeem, S., O'Neill, R. V., Paruelo, J., Raskin, R. G., Sutton, P., & van den Belt, M. (1997). The value of the world's ecosystem services and natural capital. Nature, 387, 253–260. https://doi.org/10.1038/387253a0

Khan, M. K. H., et al. (2026). Urban forests baseline and ecosystem benefits of a tropical metropolis: Case of Dhaka, Bangladesh. Scientific Reports, 16. https://doi.org/10.1038/s41598-026-46985-x

Millennium Ecosystem Assessment. (2005). Ecosystems and human well-being: Synthesis. Island Press.

Prothom Alo. (2025, June 5). 1.3 million trees in the capital, finds survey. https://en.prothomalo.com/environment/u08zh95rd5

TEEB. (2010). The economics of ecosystems and biodiversity: Mainstreaming the economics of nature — A synthesis. United Nations Environment Programme.

The Business Standard. (2025, August 5). Factories rise as forest falls: Mirsarai's troubling trade-off. https://www.tbsnews.net/bangladesh/environment/factories-rise-forest-falls-mirsarais-troubling-trade-1204691

The Climate Watch. (2025, August 12). Mirsarai Economic Zone: A trade-off between development and environment destruction. https://theclimatewatch.com/

The Daily Star. (2019, March 21). Forest land lost to govt agencies. https://www.thedailystar.net/backpage/forest-land-lost-govt-agencies-in-bangladesh-1718209

The Daily Star. (2023, July 2). Mangrove forest: Destroying the last line of defence. https://www.thedailystar.net/weekend-read/news/destroying-the-last-line-defence-3324661

Yusuf, M. (2025, August). Personal communication [Interview with senior climate journalist, Prothom Alo].